



Contents lists available at ScienceDirect

The Journal of Foot & Ankle Surgery

journal homepage: www.jfas.org

Evaluation of the Healing Status of Lateral Ankle Ligaments 6 Weeks After an Acute Ankle Sprain

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ARTICLE INFO

(Case-control observational study): 3

Keywords:

Ankle

Joint instability

Magnetic resonance imaging

Sports injuries

ABSTRACT

We aimed to investigate whether there is clinical and MRI evidence of healing of lateral ligaments 6 weeks after acute lateral ankle sprain (LAS). We prospectively enrolled 18 participants (age 32.7 ± 7.5 years) who sustained an acute LAS and underwent conservative treatment. An ankle MRI was acquired up to 48 hours and 6 weeks following the LAS. A partial tear of the anterior talofibular ligament (ATFL) was observed in 10/18 and a complete tear in 8/18 of the patients. The calcaneofibular ligament (CFL) was partially torn in 11/18 and completely torn in 1/18 of the patients. The healing status, intensity, and thickness of the ligaments, Anterior Drawer Test (ADT), and FAOS scale were assessed. A control group (CG) was composed by 17 participants (age 40 ± 13.9 years). Six weeks after the LAS, 89% of the participants presented MRI evidence of ATFL healing. The repaired ATFL was thicker in comparison with the CG ($p < .001$). The cut-off of 2.5 mm for ATFL thickness in the 6th week maximized sensitivity (62.5%) and specificity (100%). CFL and PTFL presented 94% and 100% of healing signs, respectively. In the 6th week, 11/18 (61%) participants showed mild residual instability and a mean FAOS of 80 ± 11 . The MRI revealed signs of the repair process in 89% of ATFL and 94% of CFL tears, 6 weeks after a moderate or severe LAS. The MRI findings were concomitant with enhancements in mechanical ankle stability and function.

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Lateral ankle sprains (LAS) are frequently associated with partial or complete tears of the lateral ligaments, most commonly involving the anterior talofibular ligament (ATFL), followed by injuries to the calcaneofibular ligament (CFL), and, more rarely, the posterior talofibular ligament (PTL) (1,2). The standard of care for all grades of LAS is conservative treatment, with a high rate of good outcomes. Protocols for conservative treatment include functional support and exercise strategies, with a duration of 4 to 6 weeks (1,3). This interval was empirically established based on assumptions about the healing process of the ligaments and articular stability.

This paper was carried out at Santa Helena Hospital, Brasília – DF, Brazil.

Funding Disclosure: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of interest: The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The healing process of ligament tissue is divided into 3 different phases, the inflammatory phase (~ 10 days after injury), the proliferation phase (4th-8th weeks), and the remodeling or maturation phase (up to 12 months after the injury) (4). At twelve weeks following the sectioning of the CFL in rats, the repair process displayed a sparse tissue structure, low amount of extracellular matrix, and disorganized collagen I fibers. Histological analysis suggests that the healing process, characterized by scar tissue formation in simulated grade 3 ankle sprains, could account for the poor mechanical properties of the ankle ligament. In humans, an early return to physical activities may potentially be associated with incomplete healing of the ligament tissue, resulting in ligament laxity and chronic ankle instability (4,6,7).

Magnetic resonance imaging (MRI) is a noninvasive tool that is highly accurate in identifying the ankle ligaments, and also the repair process. Imaging studies have the potential to assess the healing process in soft tissues, such as ligaments and tendons, as they are performed sequentially in longitudinal studies. However, there are limited data in the literature regarding the healing process of ankle ligaments after an acute LAS; thus, the characteristics of scar formation are not

Table 1
Demographic and baseline characteristics of the study participants

		Cases (n=18)			Control (n=17)			p-value
		Mean ± SD	CI 95%		Mean ± SD	CI 95%		
FAOS	Age	32.7 ± 7.5	28.7	36.7	40 ± 13.9	32.6	47.4	.22
	Sex	8F (44.4%)/10M (55.6%)			6F (35.3%)/11M (64.7%)			
	Weight	80.4 ± 13.7	73.1	87.7	76.1 ± 13.2	69.1	83.2	.32
	Height	1.72 ± 0.9	1.68	1.77	1.69 ± 0.1	1.64	1.75	.39
	BMI	26.8 ± 3	25.2	28.4	26.2 ± 3.1	24.6	27.9	.71
	Side	12 L (66.6%)/6 R (33.4%)			-			
	ATFL	4.44 ± 1.3	3.72	5.17	2.07 ± 0.2	1.97	2.17	<.001*
	Symptoms	76.7 ± 18.6	67.4	85.9	-	-	-	-
	Pain	84.2± 10.8	78.8	89.5	-	-	-	-
	Function	88.6 ± 13.9	81.7	95.5	-	-	-	-
	Sports	63.9 ± 26.7	50.6	77.2	-	-	-	-
	Quality of life	56.4 ± 20.2	46.3	66.4	-	-	-	-

* p-value < 0.001.

fully understood (8-10). Many previous studies have examined the healing process using indirect techniques, such as manual stress tests, stress radiographs, or arthrometry. Nevertheless, the reliability of these tests remains a subject of controversy (6). A previous study, clinically and radiologically, through MRIs, evaluated the evolution of 30 patients that had suffered an LAS. The authors reported MR indicating healing of the ATFL in 73% of the ankles, but the patients underwent the second MRI 12 weeks after the injury (10).

As the exact timeline of ankle ligament healing remains unknown, studies employing precise methods to investigate the healing process could significantly contribute to clinical decision-making and the rehabilitation process. Therefore, the aim of the current study is to investigate whether, 6 weeks after an acute LAS, conservatively treated, there is MRI and clinical evidence of the repair process in the lateral ligaments of the ankle.

Patients and Methods

Participants

The study was conducted between July 2021 and November 2022, according to the Declaration of Helsinki and approval for the project was obtained from the Local Ethics Committee (Protocol 01022818.6.0000.8093). Informed consent was obtained from all participants before data collection. Eighteen participants, with a mean age (± standard deviation) of 32.7 ± 7.5 years, with first-time acute LAS grades 2 or 3, confirmed by MRI, were included in the study. All participants were examined by an orthopedic surgeon and underwent routine ankle radiographies to rule out fractures, according to the Ottawa rules (11). Each patient underwent an ankle MRI within the first 48 hours after trauma to assess the magnitude of ligament injuries. After 6 weeks, a follow-up MRI was performed to evaluate the lateral ligament healing status.

Table 2
MRI pulse sequences and acquisition parameters for evaluation of the ankle joints

Planes	Sagittal†	Sagittal*	Axial*	Axial†	Coronal†	Coronal*	Axial*	Coronal*
Sequence	DP SPIR	T1 TSE	T1	DP SPIR	DP SPIR	T1 TSE	T1 SPIR	T1 SPIR
TR (ms)	1578	479	499	2136	2000	532	607	519
TE (ms)	40	12	10	50	50	12	15	15
Slice thickness (mm)	4	4	4	4	4	4	4	4
Interval (mm)	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
FOV (LL x AP x CC)	160×160	161×161	150×188	150×188	150×150	150×150	150×150	150×150
Voxel	0.7×0.7	0.6×0.6	0.6×0.6	0.6×0.6	0.7×0.7	0.7×0.7	0.7×0.61	0.75×0.66
Matrix	228×204	268×266	252×308	252×298	216×201	216×213	216×241	200×216
Slices	18	18	28	28	30	30	28	30
NEX / NSA	3	1.5	1	2	2.5	1	2	2
Acquisition time	02:41	02:20	02:20	02:59	02:20	02:12	02:40	02:41

Abbreviations: AP, anteroposterior; CC, craniocaudal; LL, laterolateral.

* PD weighted (proton density weighted) sequences.

† intermediate weighted.

We did not include patients with: associated injuries in the affected ankle (i.e., talar osteochondral lesions, peroneal tendons tear, or fractures in the ankle or foot); a previous history of ankle sprain injury or other lower extremity injury in the previous 6 months; a previous history of ankle fracture; a previous history of major lower limb surgery or associated pathologies (knee and ankle dysfunction, osteoarthritis); a previous history of rheumatoid arthritis, ankylosing spondylitis, diabetes, or neurological abnormalities.

Among the 18 participants, 10 presented a partial tear, and 8 a complete tear of the ATFL. There was one case of a complete tear of the CFL, 11 cases of partial tears, and 5 sprains, as well as 1 instance of a sprain of the PTFL. To provide a basis for comparison, a control group consisting of 17 participants (mean age: 40 ± 14 years) was selected based on matching BMIs and a lack of any history of ankle sprain. These participants underwent an ankle MRI to assess ATFL characteristics. Demographic and baseline characteristics of the participants in both groups are detailed in Table 1.

MRI Protocol

The participants underwent an MRI of the affected ankle within 48 hours of the acute LAS (<48h) and 6 weeks after the trauma. The MRI was performed using a 1.5-T imaging system (Ingenuia, Philips Healthcare, Best the Netherlands). Patients were examined in a supine position with the ankle in a relaxed position using a phased-array foot-and-ankle coil with 8 channels (dS HeadSpine 15 channels). The specific acquisition parameters are described in Table 2.

Imaging Evaluation

A musculoskeletal radiologist (8 years of experience) assessed the exams from both groups. A second radiologist (12 years of experience), blinded to the previous analysis, assessed the MRIs from 5 participants to check the inter-observer reliability of the ATFL thickness measurement.

MRI Assessment of the Repair Process

At the acute phase, based on fat sat intermediate weighted images with fat saturation (IW FS sequence [SPIR]), the lateral ligaments were graded as normal; sprain (grade 1); partial tear

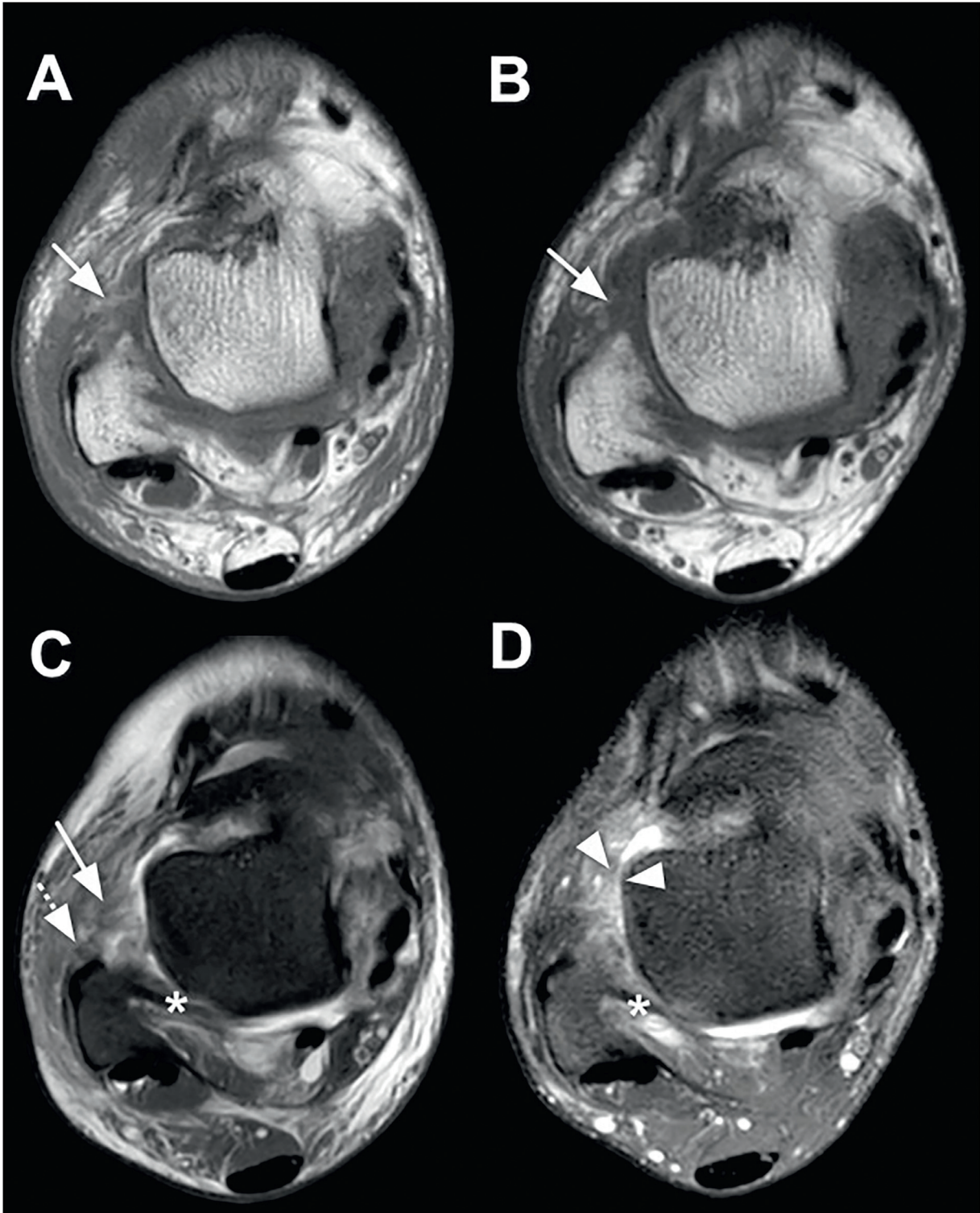


Fig. 1. Thirty-six years old woman. (A and B) Two distinct consecutive axial views, T1-weighted, at the level of the anterior talofibular ligament (ATFL), with a white arrow indicating a heterogeneous ligament with a blurred outline. (C and D) Two distinct consecutive fat sat intermediate (IW) weighted images of the same patient demonstrating a complete (grade 3) ligament tear at MRI. The dashed white arrow indicates the fibular ligament insertion, and the white arrowheads indicate the absence of fibers at the talar insertion. The white arrow at “C” indicates the ATFL midportion showing heterogeneity and discontinuous fibers. Asterisk (*) indicate the posterior talofibular ligament.

(grade 2); or complete tear (grade 3). The ligament was considered normal when it was homogeneous and had low signal intensity with uniform thickness and a normal pathway (continuous ligament fibers); Grade 1 was defined by ligament abnormality without tear;

Grade 2 was defined by a partially visible ligament due to fiber discontinuities, irregularities, or a heterogeneous signal increase; and Grade 3 was characterized by nonvisualization and/or complete interruption of the ligament (Fig. 1) (12–16).

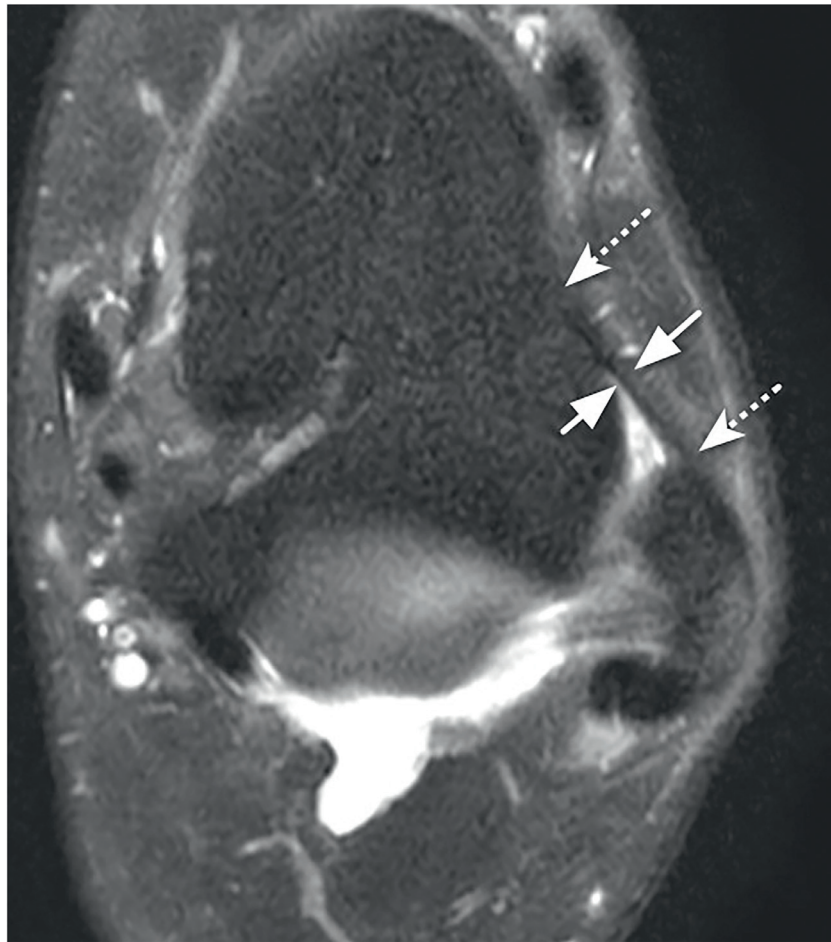


Fig. 2. Fat sat intermediate (IW) weighted axial section, at the level of the anterior talofibular ligament (ATFL), with dashed white arrows indicating the fibular and talar insertions, and a white solid arrow indicating the ligament midportion, where the ligament thickness was measured. The ends of the white arrows indicate the superficial and deep ligamentous surfaces used for measurement.

After 6 weeks, the participants underwent a new MRI to reassess and classify the lateral ligaments, as either; healed, when they displayed initial signs of healing, characterized by the presence of fibrous scar tissue, which was designated as a ligament with either a thin or thick appearance, an irregular and wavy contour, and abnormal signal intensity, or nonhealed, when there was a complete interruption, and the fibers could not be visualized (12–16).

The signal intensity was graded as a) very low, when the signal intensity was similar to tendons; b) low and homogeneous, when the signal intensity was uniform between tendon and muscle (expected normal signal); c) low and heterogeneous, when signal intensity was nonuniform between tendon and muscle; d) intermediate, when the signal intensity was similar to muscle; or e) high, when the signal intensity was greater than that of the muscle (12–16) (Figs. 2, 3, and 4).

ATFL Thickness Measurement

When the ATFL was identified, its thickness was measured using the 2D SPIR axial sequence (spectral presaturation with inversion recovery). This sequence is considered optimal for representing the structure of the ligament because it employs a combination of different fat suppression techniques (14). Measurements were taken 3 times at the midpoint of the ligament, and the mean value in millimeters was recorded for subsequent comparisons (Fig. 5) (14).

Conservative Treatment

After being enrolled in the study, all participants underwent a conservative treatment regimen in accordance with the institution's standard care protocol, lasting for 6 weeks. This protocol entailed 1 to 2 weeks of removable immobilization on the affected limb, with weight-bearing permitted based on individual tolerance. Physiotherapy commenced after the first week and included analgesic therapies, kinesiotherapy, proprioception exercises, and muscle-strengthening exercises (3).

Outcomes

The injured ankle was clinically assessed during the acute phase (within 48 hours after the trauma) and at the 6-week follow-up, by a single ankle surgeon with 12 years of experience. The outcomes assessed were the Foot and Ankle Outcome Score (FAOS) and Anterior Drawer Test (ADT). The FAOS and its subscales were used to quantify self-reported function and symptoms in 5 separate subscales: Pain, Other Symptoms, Activities of Daily Living (ADL), Sport and Recreation Function, and Foot and Ankle-related Quality of Life. A percentage score from 0 to 100 is calculated for each subscale, with 100 representing the best possible score (17,18). Mechanical ankle instability was assessed using the ADT, conducted with participants seated on a plinth with the knee flexed and the tested limb hanging freely above the ground. The lower leg was stabilized with the foot at approximately 20° of ankle plantarflexion. The examiner held the posterior-inferior aspect of the calcaneus and applied a posterior-anterior force to "draw the talus forward in the ankle mortise." The test was initially performed on the contralateral side to assess the normal translation of the ankle and compare it with the affected side. To determine the magnitude of the mechanical instability, the test was quantified from 0 to 4+, where 4+ represents gross instability of the ankle (19).

Statistical Analysis

Descriptive statistics were used to present the study population. Continuous variables are presented as mean $\pm$ SD or median, and Confidence Interval of 95% and analyzed using the Mann-Whitney test. To determine cut-off values of statistically significant quantitative criteria, receiver operating characteristic (ROC) analysis was used to calculate sensitivity and specificity. Inter-observer agreements were evaluated using the interclass correlation coefficient (ICC), interpreted as follows: less than 0.2, slight; between 0.2 and 0.4, fair; between 0.4 and 0.6, moderate; between 0.6 and 0.8, substantial; and greater than 0.8, almost perfect. *p*-values less than .05 indicated statistical significance. All statistical analyses were performed using SPSS version 25 (IBM Corp., Armonk, NY, USA).

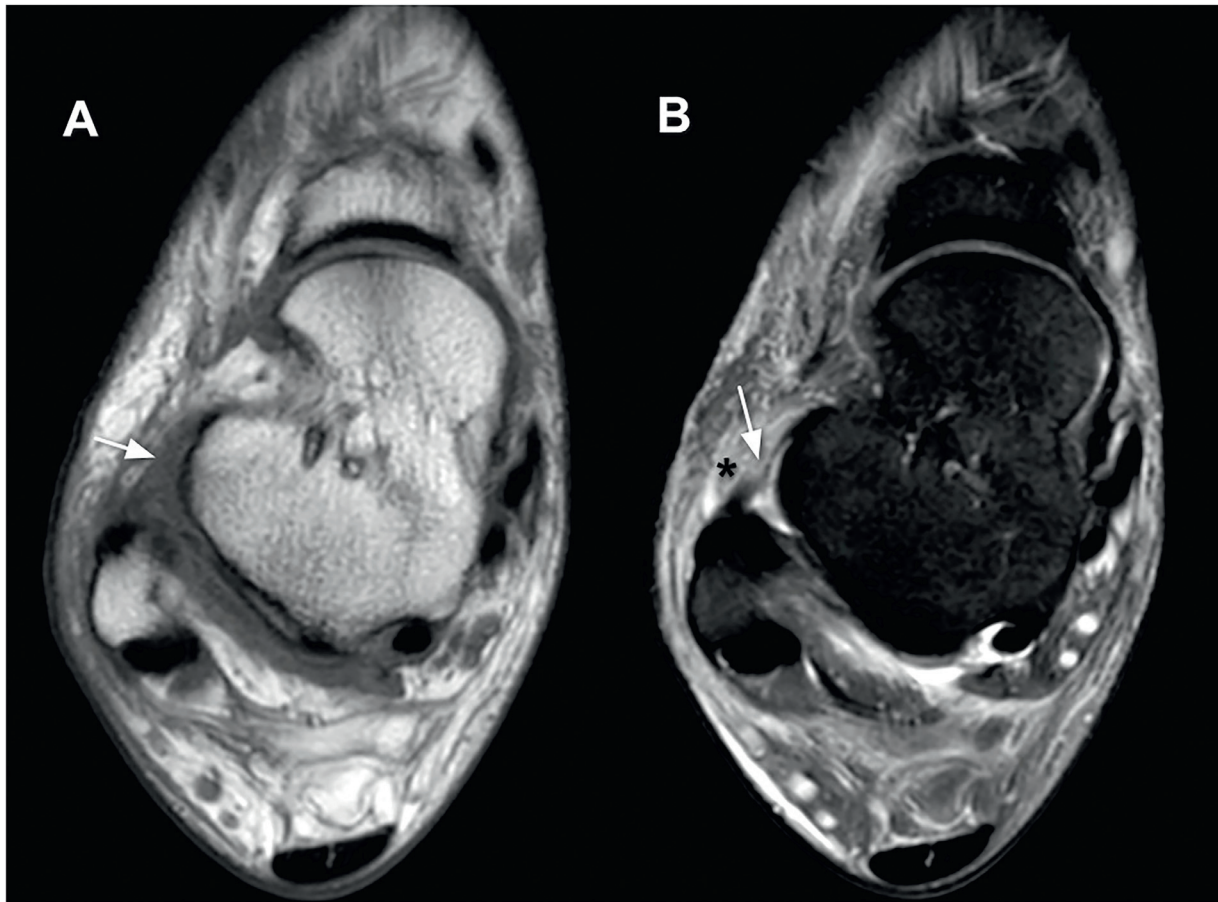


Fig. 3. Grade 2 (partial) ligament tear. “A” shows a T1-weighted axial section, with the anterior talofibular ligament (ATFL) indicated by the white arrow. “B” represents a fat sat intermediate (IW) weighted image at the same anatomical level, with the arrow indicating partial discontinuity of the fibers, with fluid signal partially replacing the normal ligamentous signal intensity.

Results

After 6 weeks, 16 out of the 18 (89%) participants exhibited signs of ATFL healing on the MRI, while 2 participants showed nonhealing (both experienced acute complete tears). Among the participants with signs of healing, 7/16 (43.8%) demonstrated intermediate signal intensity, and 6/16 (37.5%) displayed low and heterogeneous signal intensity on the fluid-sensitive fat saturation sequence MRI (Table 3). The median thickness of the ATFL repaired with fibrous tissue was 4.3 mm (range from 2.6 to 6.2 mm) in the LAS group; while in the control group, the values was 2 mm (range from 1.8 to 2.4 mm) ($p < .001$). Receiver operating characteristic (ROC) analysis demonstrated a significant curve (AUC = 0.768, EP = 0.08; $p < .01$; 95% CI = 0.594–0.942). The cut-off point that maximized sensitivity and specificity was 2.5 mm of ATFL thickness, with a sensitivity of 0.625 and specificity of 1.0 (Fig. 6). The interobserver reliability for the measurement of ATFL thickness was considered nearly perfect, with an ICC value of 0.998 (95% CI: 0.985 – 1.00).

Regarding the CFL, there was 1 case of a complete tear of the CFL, 11 cases of partial tears, and 5 sprains, in the acute phase MRI. After 6 weeks, 16 of the 17 (94.1%) injured CFL presented initial signs of healing, with the formation of immature scar tissue, showing intermediate (58.8%) or high (23.5%) intensity signals. Lastly, 1 participant (5.6%) suffered a PFTL sprain (grade 1), with signs of healing, presenting a low and heterogeneous signal intensity after 6 weeks (Table 4).

At the acute phase, 17 of the 18 (94%) participants presented gross mechanical instability (ADT with 3 or 4 “crosses”). After 6

weeks, 11 participants (61%) showed mild residual instability, classified as “1 cross”, and the remainder (39%) presented no mechanical instability. At the 6th week, the mean FAOS score was 80 ± 11 , with the subscales varying from 88 ± 14 to 56 ± 20 , respectively, for Function, daily living, and Function, sports, and recreational activities.

Discussion

The main findings indicate that, 6 weeks after conservative treatment for acute moderate or severe LAS, most participants displayed signs of healing in the lateral ligaments, as observed on the MRI. Immature scar tissue was observed in 16/18 (89%) and 16/17 (94.1%) of the participants with injury of the ATFL and CFL, respectively. The mean thickness of the ATFL was 4.22 ± 1.6 mm, significantly thicker compared to the control group. Furthermore, in the 6th week, most of the participants presented no or mild mechanical instability, with good ankle function, despite residual limitations, especially in daily and sports activities. This novel investigation may contribute to the current knowledge about the healing of lateral ligaments after an acute ankle sprain, providing a basic understanding of the ligament healing cascade and the rehabilitation process.

Ligament strength has been related to the number, size, and quality of collagen fibers (5). An initial LAS results in stretching or disruption of the collagen fibers of the lateral ligaments, causing structural tissue damage. Histological studies demonstrated that biological ligament

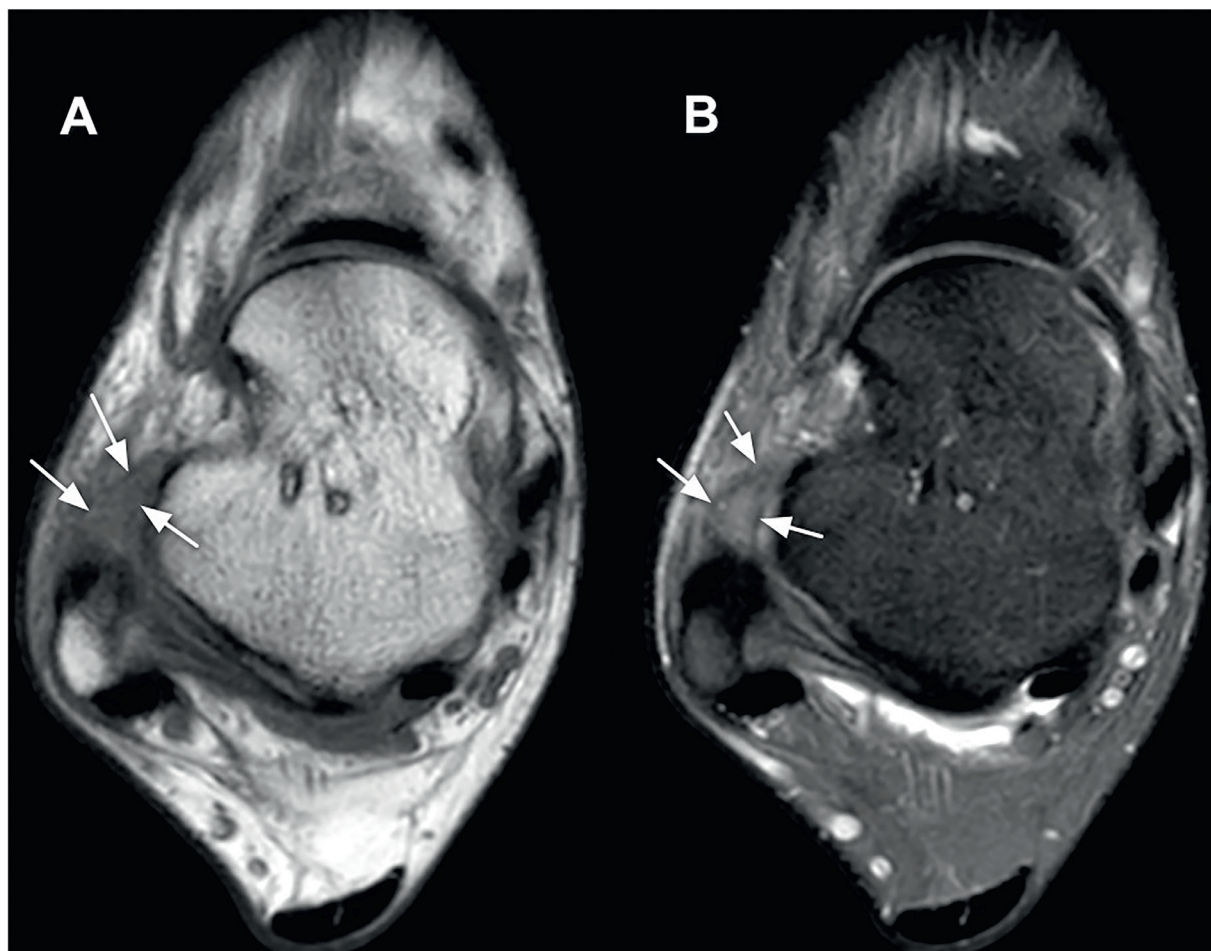


Fig. 4. Follow-up MRI of the same case of Fig. 3, illustrating the repairing fibrocartilaginous process, in this specific case with ligament thickening. “A” shows a T1-weighted axial section and “B” a fat sat intermediate (IW) weighted image at the level of the anterior talofibular ligament (ATFL). White arrows indicate the ATFL showing a thickened ligament with heterogeneous signal intensity, predominantly with intermediate signal intensity.

healing can be divided into 3 different phases: (I) the inflammatory phase (until 10 days after the trauma), (II) the proliferation phase (4th–8th weeks), and (III) the remodeling or maturation phase (up to 1 year after the trauma). During the proliferation phase, the tissue responds with vascular ingrowth, fibroblast proliferation, and new collagen formation (4). As the healing process progresses over time, scar tissue fills in and surrounds the torn portion, bridging the tear (20). During the period of maturation of the collagen fibers, the structure, organization, biomechanical properties, and proprioceptive sensation may not be fully restored due to the development of scar tissue. A very low extracellular matrix and disorganized collagen fibers characterize this tissue, which may explain the loss of mechanical competence (5).

The healing process of ankle ligaments after an LAS has not been completely investigated, especially through diagnostic imaging techniques (9,10). Most previous studies analyzed ligament healing indirectly through manual tests, stress radiographs, or arthrography (9); however, these tests show insufficient accuracy for specific ligament involvement (9,21). In the present study, we assessed ligament healing using an MRI and a manual test (ADT). Six weeks after a moderate or severe injury, only 2 ATFL (11%) and 1 CFL (5.6%) showed no signs of healing on the MRI. In all nonhealed cases, the initial injury involved a complete tear of the ligament. These findings align with those of previous studies (9,10). De Simoni et al. (10) reported MRI evidence of healing in 73% and 92% of the ankles with injury of the ATFL and CFL, respectively, 12

weeks after an ankle sprain. However, the study included 30 participants, 8 with a previous ankle sprain. In addition, the sample included patients who had suffered the LAS for up to 19 days. Another author (9) retrospectively reviewed the ligament healing process over time. The study included 84 individuals with a history of ligamentous injury over a 5-year period, who underwent an MRI to analyze the ATFL morphology. Within 2 weeks of the injury, half of the patients presented an ATFL defect and after 7 weeks, nonhealing was observed in 11%. However, despite the large sample size and MRI evaluation, the study included immature patients (aged from 13 to 67 years), and did not assess clinical information, such as a previous sprain, the treatment performed, or CAI.

The increased translation of the talus within the tibiofibular mortise and positive anterior drawer test (ADT) can be observed after an acute LAS or in the CAI context (4,6,7). In the present study, 11 / 18 participants evolved with residual mild (1+/4+) (61%) and 7 / 11 (39%) with no mechanical instability, according to the ADT assessment. Previous studies (22,23) assessed ligament injury and healing by performing the ADT assessment. The authors reported approximately 10% of positive ADT 6 (22) to 8 (23) weeks after the LAS. These conflicting results can be attributed to the absence of gradation (i.e., “crosses”) or a precise definition of the results. In addition, the authors (22,23) only categorized the results as either positive or negative, and the subjectivity of the manual tests may also have contributed to the discrepancies. Thus, tests were

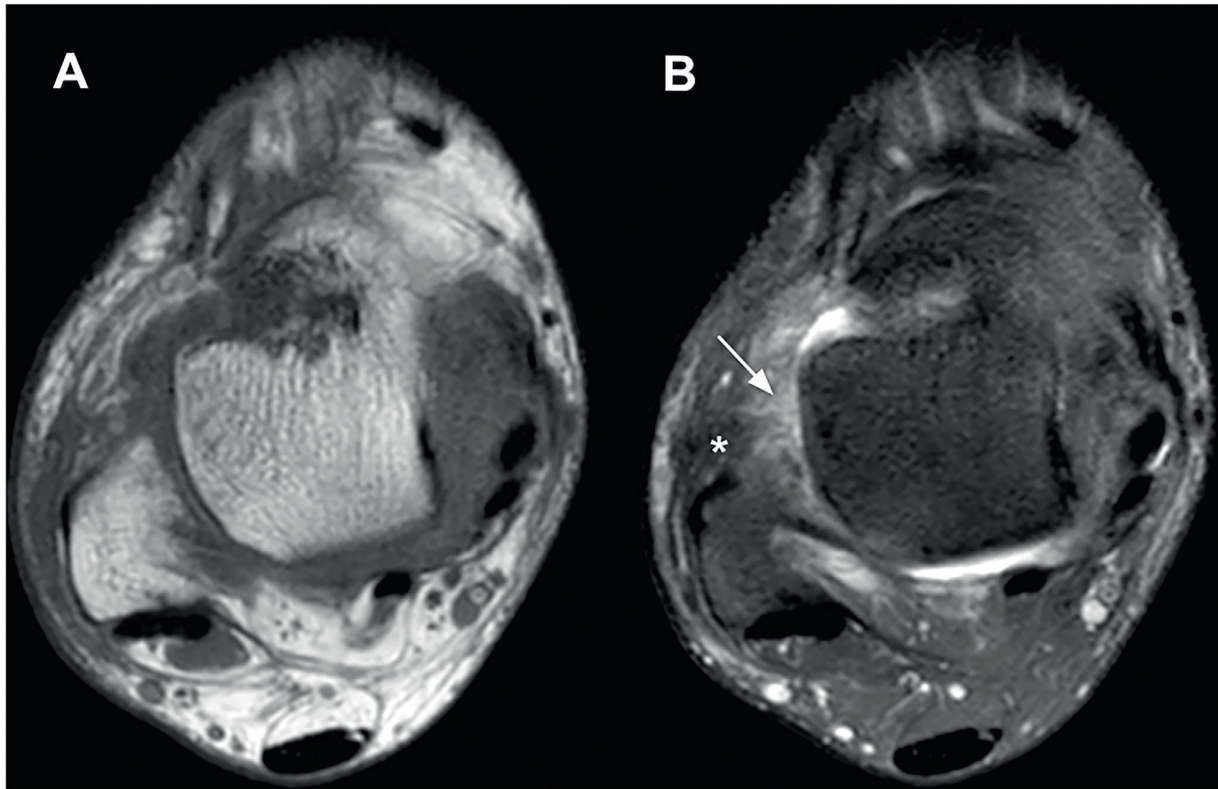


Fig. 5. Follow-up MRI of the same patient in Fig. 1, illustrating the lack of adequate repair, without fibroscar tissue at the distal insertion (white arrow).

considered positive when there was gross instability of the ankle, which was not observed in our sample 6 weeks following the injury. The residual mechanical instability, as assessed through the ADT, may be associated with the maturation of fibrous scar tissues and the organization of collagen fibers.

The ATFL anatomy has been extensively studied. A systematic review (24) assessed the anatomy of the ATFL, reporting a mean thickness varying from 5 to 11.1 mm; however, most of the studies measured the ligament of cadavers with different methodologies. An imaging study (25) compared the morphological characteristics of lateral ankle ligaments using 3D MRI and found a mean thickness in the central portion of the ATFL in healthy individuals of 4.0 ± 1.0 mm. In our study, the ATFL was significantly thicker in the LAS group. The injured participants presented a median thickness of 4.3 mm (range

from 2.6 to 6.2 mm), while in the control group, the values were 2 mm (range from 1.8 to 2.4 mm). In addition, we observed a cut-off point of 2.5 mm, a thickness value that may represent a healed ATFL. The injured ligament undergoes repetitive stress while attempting to stabilize the joint, due to the altered biochemical and mechanical properties, which may cause stretch at the fibrous scar (9). Furthermore, the excess of weaker type III collagen formation during the remodeling phase can contribute to chronic elongation of the ligament, with subsequent laxity and joint mechanical instability and/or joint impingement (Antero-lateral Ankle Impingement Syndrome) (9). The mechanical instability may cause chronic ankle instability (CAI) and residual symptoms (6,7).

Clinically, the disruption of the ATFL is associated with pain, instability, and functional impairments. Several studies have demonstrated that after an acute LAS, a high percentage of individuals experience residual symptoms (3,4,6,7). We observed a decrease in ankle function, with a mean FAOS of 76.7 ± 18.6 . Moreover, analyzing the subscales, there is a greater decline in sports and recreational activities and overall quality of life, with scores of 63.9 ± 26.7 and 56.4 ± 20.2 , respectively. Our results are consistent with previously published studies (3,26). There are 2 main reasons which can explain these findings; first, although ligament healing occurs after from 6 to 12 weeks, the final maturation and remodeling can last much longer, up to 12 months (9,27). Second, besides the function of mechanical stabilization of the ankle provided by the ligaments, they have a crucial function in joint proprioception. Histological studies show that the ligaments contain mechanoreceptors with nerve endings, providing kinesthesia and statesthesia that cause activation or inhibition of muscular activities, through neurological feedback (11,28). Furthermore, a previous study demonstrated that 6 weeks after an acute LAS, there is significant atrophy of the muscles and tendons around the ankle, accompanied by functional impairments (29). Therefore, residual functional

Table 3
Anterior talofibular ligament (ATFL) MRI 6 weeks after an acute ankle sprain

	Group	
	Cases (n = 18)	Control (n = 17)
Intensity	(%)	(%)
Very low	5.6	
Low and homogeneous		100
Low and heterogeneous	33.3	
Intermediate	38.9	
High	11.1	
Nondefined	11.1	
Result		
Normal		100
Fibrous scar tissue	88.9	
Nonhealed	11.1	

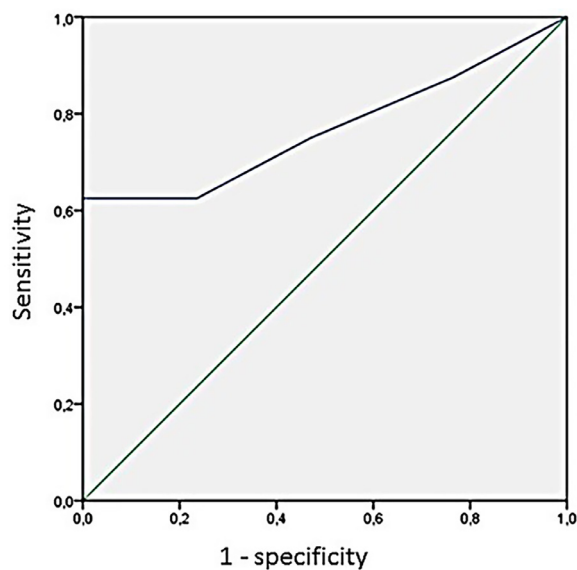


Fig. 6. ROC curve evaluating the sensitivity and specificity of the thickness of the Anterior Talofibular Ligament (LTFA).

Table 4
Qualitative assessment of the calcaneofibular ligament (CFL) and the posterior talofibular ligament (PTFL) <48 h and 6 weeks after the ankle sprain (n = 18)

Ligament	Ligament status <48 h after injury			
	No lesion	Sprain	Partial tear	Complete tear
CFL	1 (5.6%)	5 (27.8%)	11 (61%)	1 (5.6%)
PTFL	17 (94.4%)	1 (5.6%)	—	—
Healing status 6 weeks after injury				
CFL	Normal	Healed 100%	Healed 100%	Nonhealed 100%
PTFL	Normal	Healed 100%	—	—

Note: We classified cases as 'healed' when the ligament displayed initial signs of healing, characterized by the presence of fibrous scar tissue, and as 'nonhealed' when there was a complete interruption and the fibers could not be visualized.

impairments may persist alongside the signs of ligament healing, due to a lack of proprioception and muscle atrophy. Taken together, despite the indirect MRI signs of ligament healing 6 weeks after the LAS, caution must be exercised throughout the remodeling process in rehabilitation to prevent an early return to activities (6).

We are aware of several limitations of our study. First, this study enrolled a relatively small number of participants. One of the barriers to achieving a larger sample size is the prospective design, which involved the inclusion of only participants with first-time moderate or severe LAS that occurred within up to 48 hours. Additionally, we did not account for the presence of multiple bands in the ATFL or evaluate the medial complex. Furthermore, mechanical instability was assessed solely through a subjective test (ADT); however, all participants were examined by a single physician. Furthermore, since this was an observational study, there was no control over whether patients adhered to the conservative treatment protocol. Lastly, our follow-up period extended only to 6 weeks; thus, we acknowledge that our data limit the interpretation of our conclusions to the short term.

In conclusion, in the present study, the MRI revealed signs of healing of the lateral ligaments of the ankle in 16/18 (89%) for ATFL and 16/17 (94.1%) for CFL of the participants, 6 weeks after experiencing a moderate or severe acute LAS, which was treated conservatively. Furthermore, these MRI findings were accompanied by improvements in mechanical

ankle stability and satisfactory functional outcomes, despite residual impairments, particularly in sports activities.

Ethics Statement

The study was conducted according to the Declaration of Helsinki and approval for the project was obtained from the Local Ethics Committee (Protocol 01022818.6.0000.8093).

Patient Consent

Informed consent was obtained from all participants before data collection.

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